

Saurabh Kadekodi

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Senior systems researcher with 5+ years of experience in improving the efficiency and reliability of exascale distributed storage systems.

EDUCATION	Carnegie Mellon University 2014 - 2020 <i>Ph.D. in Computer Science</i> Advisors: <u>Prof. Gregory Ganger</u> , <u>Prof. Rashmi Vinayak</u> and <u>Prof. Garth Gibson</u> Thesis: Disk-Adaptive REdundancy (DARE) : tailoring data redundancy to disk-reliability heterogeneity in cluster storage systems (<i>dissertation</i>) Northwestern University 2012 - 2013 <i>Masters in Computer Science</i> Advisor: <u>Prof. Peter Dinda</u> Pune Institute of Computer Technology (University of Pune) 2005 - 2009 <i>Bachelors in Computer Engineering</i>
RESEARCH INTERESTS	Exascale Distributed Systems, Fault Tolerance, Data Layout and Management, File and Storage Systems, Benchmarking, ML for Storage, Storage for ML
CURRENT RESEARCH FOCUS	Redesigning reliability in exascale storage clusters Conceptualized Disk-Adaptive REdundancy (DARE), in which data redundancy is dynamically tailored to observed disk failure rate heterogeneity. Analyzed over 5.3 million disks spanning 60+ makes/models from production environments of Google, NetApp, and Backblaze. Designed three DARE systems: <i>HeART</i> , <i>Pacemaker</i> and <i>Tiger</i> which provide up to 20% space-savings (worth multi-million dollars) in large-scale storage clusters while never compromising data safety. Currently leading a project to design and implement DARE in Google's cluster storage system. ML for storage device emulation Designing a high-accuracy, low-latency storage device emulator that can be used in place of software storage device emulators. An accurate ML model can enable research in various ways: a) conduct what-if experiments without spending thousands on infrastructure, b) allowing storage manufacturers and companies to share a performance-sensitive emulator with academia without violating confidentiality.
AWARDS / GRANTS	Best paper award for FastCommits in ATC 2024. Google Innovation Fund award for Disk-Adaptive Redundancy (DARE) in Sep 2023. DARE was one of 6 out of 200 submissions that was awarded a 4 person-year funding to implement DARE in the Colossus distributed file system. Helped write a successful NSF grant proposal that received federal funding of \$986,805 (#1956271) on understanding and addressing device-reliability heterogeneity in large-scale distributed storage systems. Nominated by CMU for ACM SIGOPS Best Dissertation Award 2021 for Ph.D. thesis on Disk-Adaptive Redundancy. Best Alumni Research Award 2010 PICT for Space Maps in Ext4. USENIX travel grant - FAST 2015, FAST 2016, FAST 2018, FAST 2019. NSF travel grant - ACM SOSP 2020.
CURRENT WORK	Google - Senior Research Scientist Jan 2021 - present Designing analytical/simulation frameworks, exploring machine learning methods, and drawing insights from massive data sets across various storage systems at Google. Provide critical inputs to inform both short-term tactical actions and long-term technology choices and strategies to help better understand Google's workload and optimize storage design in an increasingly heterogeneous fleet.

TCO-driven Storage Provisioning for Exascale Data Centers

Kim Timothy, [Kadekodi Saurabh](#), Merchant Arif, Nema Prashanth, Menon Jai, Rashmi K. V., Ganger Gregory R. *Eurosys 2026 (to appear)*

Okapi: Decoupling Data Striping and Redundancy Grouping in Cluster File Systems

Athlur Sanjith, Kim Timothy, [Kadekodi Saurabh](#), Maturana Francisco, Ramos Xavier, Merchant Arif, Rashmi K. V., Ganger Gregory R. *OSDI 2025*

Morph: Efficient file-lifetime redundancy management for cluster file systems

Kim Timothy, Athlur Sanjith, [Kadekodi Saurabh](#), Maturana Francisco, Delvira Dax, Merchant Arif, Rashmi K. V., Ganger Gregory R. *SOSP 2024*

FastCommit: resource-efficient, performant and cost-effective file system journaling

Shirwadkar Harshad [Kadekodi Saurabh](#), Tso Theodore. ***Awarded Best Paper*** *ATC 2024*

Thesios: Synthesizing Accurate Counterfactual I/O Traces from I/O Samples

Phothilimthana Pitchaya Mangpo* [Kadekodi Saurabh](#)*, Ghodrati Soroush, Moon Selene, and Maas Martin. *(blog, traces)* *ASPLOS 2024*

Practical Design Considerations for Wide Locally Recoverable Codes (LRCs)

[Kadekodi Saurabh](#), Silas Shashwat, Clausen David, and Merchant Arif. ***Among top picks fast-tracked to ACM TOS*** *ACM TOS 2023, FAST 2023*

Tiger: disk-adaptive redundancy without placement restrictions

[Kadekodi Saurabh](#), Maturana Francisco, Athlur Sanjith, Merchant Arif, Rashmi K. V., and Ganger Gregory R. *OSDI 2022*

WineFS: a hugepage-aware file system for persistent memory that ages gracefully

[Kadekodi Rohan](#), [Kadekodi Saurabh](#), Ponnappalli Soujanya, Shirwadkar Harshad, Ganger Gregory R., Kolli Asheesh, Chidambaram Vijay *SOSP 2021*

Pacemaker: Avoiding HeART attacks in storage clusters with disk-adaptive redundancy

[Kadekodi Saurabh](#), Maturana Francisco, Suhas Jayaram Subramanya, Yang Junchen, Rashmi K. V. and Ganger Gregory R. ***Highlights presentation in ACM Systor 2022.*** *(arXiv)* *OSDI 2020*

Cluster storage systems gotta have HeART: improving storage efficiency by exploiting disk-reliability heterogeneity

[Kadekodi Saurabh](#), Rashmi K. V. and Ganger Gregory R. *(press)* *FAST 2019*

Geriatric: Aging what you see and what you don't see

[Kadekodi Saurabh](#), Nagarajan Vaishnavh, Ganger Gregory R. and Gibson Garth A. *ATC 2018*

A case for packing and indexing in cloud file systems

[Kadekodi Saurabh](#), Fan Bin, Madan Adit, Gibson Garth A. and Ganger Gregory R. *HotCloud 2018*

Software-Defined Storage for Fast Trajectory Queries using a DeltaFS Indexed Massive Directory

Zheng Qing, Amvrosiadis George, [Kadekodi Saurabh](#), Kuchnik Michael, Cranor Chuck, Gibson Garth A., Settlemeyer Brad, Grider Gary, and Guo Fan *PDSW 2017*

Caveat-Scriptor: Write Anywhere Shingled Disks

[Kadekodi Saurabh](#), Pimpale Swapnil and Gibson Garth A. *HotStorage 2015*

Space Maps in Ext4

[Kadekodi Saurabh](#) and Jain Shweta. To be integrated in the mainline Linux kernel. *OLS 2010*

Compression in Checkpointing and Fault-Tolerant Systems

[Kadekodi Saurabh](#) *Technical Report*

VM Co-Migration

[Kadekodi Saurabh](#), Shi Chao, Zhang Qingyuan *Technical Report*

NIC of Time: smart NIC offloading

Amos Brandon, [Kadekodi Saurabh](#), Kiyak Serhat *Technical Report*

Active Databases

Avci Besim, [Kadekodi Saurabh](#), Paul Arindam *Technical Report*

DNA Compression

[Kadekodi Saurabh](#) *Technical Report*

RESEARCH EXPERIENCE	Google - <i>Student Researcher</i>	<i>Summer 2019 - Dec 2020</i>
	Analyzed disk failure and reliability data for Google's cluster storage systems.	
	NetApp - <i>Research Intern</i>	<i>Spring 2019</i>
	Analyzed disk failure and reliability data for NetApp's enterprise storage systems spanning over 4.3 million HDDs over 60 makes/models.	
	Alluxio - <i>PhD Intern</i>	<i>Summer 2017</i>
	Designed and developed a packing and indexing layer in the writeback cache of Alluxio (a distributed user-level file system) to transform arbitrary user workloads to write patterns that are better suited for the cloud. My optimization resulted in $>60\times$ increase in throughput and $>25000\times$ reduction in cost required to ingest data into Amazon S3.	
	Google - <i>PhD Intern</i>	<i>Summer 2016</i>
	Worked at the intersection of the Storage Analytics and Colossus (distributed file system) teams on analyzing access patterns from Google services and designing a load-balancing strategy to avoid disk hot-spots in data centers with over a hundred thousand disks with heterogeneous capacities and speeds.	
	Apple - <i>PhD Intern</i>	<i>Summer 2015</i>
	Spent a summer as a part of the file systems team in the CoreOS division.	
	University of Wisconsin - Madison - <i>Graduate Research Intern</i>	<i>Summer 2014</i>
	Advisor: Prof. Remzi Arpaci-Dusseau	
	SSD over Infiniband	
	This study compared the performance between a locally connected SSD and a remotely connected SSD (over Infiniband). Using the lightweight SCSI RDMA protocol (SRP) for communication, we analyzed the costs of accessing remote SSDs and gained insights into enhancing software architectures of next-gen data centers from the storage perspective.	
	Price of Ext4	
	Measured the software overhead of the Ext4 file system with the advent of storage devices with microsecond latencies. We threw light on simple optimizations that can be done to remove the bottlenecks observed in sub-modules of Ext4.	
	Soft Corner - <i>Software Developer</i>	<i>Oct 2010 - Sep 2012</i>
	PatientScribe	
WORK EXPERIENCE	Worked on data reliability, fault tolerance, and audit trails at the clinical level. Developed global data reliability using data analytics. Also developed a distributed data-storage algorithm for seamless remote backup and restore.	
	Project-X	
	Designed and developed a custom stackable context-based framework for the social network. Developed a suggester module based on a graph-based traversal of the user connection web to help generate meaningful suggestions based on connected entities. Extended WWW SQL Designer by adding collaboration, versioning, and a chat module to build a collaborative real-time block diagram and discussion tool.	
	Spring Computing Pvt. Ltd. - <i>Software Engineer</i>	<i>Jun 2009 - Oct 2010</i>
	Emdebian Research	
	Created a 9 MB JFFS2 filesystem image using Emdebian Crush (an online repository that helps create BusyBox-based root filesystem images) and ported the resulting filesystem image on an ARM board.	
	Kernel Porting on Embedded Devices	
	Participated in the full cycle from porting Linux kernel 2.6.30 onto 3 customized ARM boards, testing and tweaking peripherals to adapt to the new kernel and making public releases of the boards.	
	Teaching Assistant for <u>Distributed Systems (CMU 15-640)</u>	<i>Fall 2016</i>
	Lead Teaching Assistant for <u>Storage Systems (CMU 15-746)</u>	<i>Fall 2015</i>
	Mentored an undergraduate student through the NSF REU program for underrepresented communities.	<i>Summer 2022</i>
	Talk at Shelter Associates on computer science and storage systems research for healthcare and sanitation field workers and administrative staff.	<i>Dec 2021</i>
TEACHING		
OUTREACH		

LEADERSHIP	Leading project to design and implement DARE at Google - Leading a team of 4 people to realize DARE in exascale storage clusters at Google. <i>September 2023 - present</i>
	Mentored 16 undergraduate and 16 Masters students' capstone projects - across 4 areas: building social networks, shingled disk layouts, optimizing for small files in cloud file systems, and incorporating reliability heterogeneity awareness in HDFS. <i>Fall 2010 - present</i>
	Co-Hosted Ph.D. research intern - at Google. Built the Quilt system to synthesize realistic disk IO traces by stitching sampled IO traces. <i>Summer 2022</i>
	Mentored 2 junior Ph.D. students - who were in their first year while I was a senior Ph.D. student, and continued mentorship after my graduation. <i>Spring 2020 - present</i>
	Product Owner / Team Lead of Project-X - led a team of engineers, who in conjunction with a team of undergraduate students built a social network from scratch. Project-X ran successfully in India for about 5 years with a few thousand users and provided a platform for faculty, students, and industry to connect via a social network of undergraduate capstone projects. <i>2010 - 2012</i>
	President of the PICT Art Circle - led the undergraduate drama society and expanded it from 30 to 50 members over two years. <i>2007 - 2009</i>
	Mentored Teaching Assistants - for two iterations of the storage systems course. <i>2016 - 2017</i>
	President of SPICMACAY-CMU - led the <i>Society for Promoting Indian Classical Music And Culture Among Youth</i> (CMU student branch) and organized two concerts of internationally acclaimed Indian classical music artists. <i>May 2019 - May 2020</i>
	Event Chair (Drama) and elected board member of CMU's Indian Graduate Student Association (IGSA) - established a 30-person team of theater enthusiasts at CMU and performed two one-act plays translated from Marathi to English. <i>2016 - 2018</i>
INVITED TALKS	Talk at Databricks on Thesios <i>Mar 2025</i>
	Guest lecture at UC Irvine (UCI CS230) on <i>Redundancy management in exascale storage systems</i> in the Distributed Computer Systems course <i>Spring 2025</i>
	Talk at Texas A&M in the Computer Systems Seminar series <i>Fall 2024</i>
	Talk at joint USENIX OSDI / ATC providing an expert overview of selected conference papers in storage systems <i>Jul 2021</i>
	Guest lecture at Rutgers University on <i>Memory & Computers: making computational memory architectures accessible to a non-specialist audience in the Pasts and Presents: Recollection, Archives, and Curation</i> seminar course. <i>Fall 2018, Spring 2020, Fall 2021</i>
	Talk at CMU PDL in the summer invited talk series <i>Jul 2021, Jul 2023</i>
	Talks at Google on <i>Pacemaker</i> in the D, Colossus and Storage Analytics teams <i>Aug 2020</i>
	Talks at Google on <i>HeART</i> in D, Colossus and Storage Analytics teams on HeART <i>Jan 2019</i>
	Talk at Microsoft Research (India) on <i>HeART</i> and disk-adaptive redundancy <i>Dec 2018</i>
	Talk at Google on <i>Geriatric</i> in D, Colossus teams <i>Jul 2018</i>
	Guest lecture at CMU on <i>Packing and Indexing in Cloud File Systems</i> in the Advanced Cloud Computing course <i>Spring 2018</i>
	Guest lecture at CMU on <i>GFS</i> in the Distributed Systems course <i>Fall 2016</i>
	Guest lecture at CMU on <i>Caveat Scriptor</i> in the Storage Systems course <i>Fall 2015</i>
CONFERENCE / WORKSHOP PRESENTATIONS	Co-presented 3+ hour Tutorial at ISIT 2024 providing a systems perspective of modern coding theory applications to exascale storage clusters.
	Presented Practical Wide LRCs at FAST 2023 <i>Feb 2023</i>
	Presented Tiger at OSDI 2022 <i>Jul 2022</i>
	Highlights presentation of Pacemaker at ACM Systor 2022 originally presented at USENIX OSDI 2020. <i>Jun 2022</i>

	Presented Pacemaker at OSDI 2020	<i>Nov 2020</i>
	Presented HeART at the 1st <u>Toronto Systems Workshop</u> at University of Toronto	<i>Oct 2019</i>
	Presented HeART at FAST 2019	<i>Feb 2019</i>
	Presented Geriatrix at ATC 2018	<i>Jul 2022</i>
	Presented Packing and Indexing in Alluxio at HotCloud 2018	<i>Jul 2018</i>
	Presented Caveat-Scriptor at HotStorage 2015	<i>Jul 2015</i>
	Presented Space Maps at OLS 2010	<i>Jul 2010</i>
SERVICE	Program Committee for USENIX File and Storage Technology Conference.	<i>FAST 2025, 2027</i>
	Associate Editor for Communications of the ACM Practice Section.	<i>CACM 2025</i>
	Program Committee for NeurIPS Workshop on Machine Learning and Systems.	<i>MLForSys 2025</i>
	Program Committee for Conference on Machine Learning and Systems.	<i>MLSys 2025</i>
	Program Committee for ACM Doctoral Systems Workshop.	<i>ACM SysDW 2024</i>
	Program Committee for ACM European Conference on Systems.	<i>ACM Eurosys 2023</i>
	External reviewer for ACM Transaction on Storage.	<i>ACM TOS 2022</i>
	External reviewer for ACM Transaction on Computer Systems.	<i>ACM TOCS 2021</i>
	External reviewer for IEEE Transaction on Computers.	<i>IEEE TC 2021</i>
	Area reviewer for the accepted papers in storage systems	<i>OSDI / ATC 2021</i>
	Part of the editorial board for the storage systems area of <u>JSys</u> , the first diamond open-access journal covering all areas of computer systems research.	<i>JSys 2021</i>
	Student reviewer as a part of the shadow program committee	<i>EuroSys 2020</i>
OPEN SOURCED CODE	Geriatrix – an aging framework for file systems.	<u>Github</u>
	WineFS – a huge-page friendly, age-resistant persistent memory file system.	<u>Github</u>
	NIC of time – a smart NIC offloading framework.	<u>Github</u>
	Peloton GC – garbage collection for an in-memory database.	<u>Github</u>
STUDENTS MENTORED	Undergraduate Students – Harshad Shirwadkar, Aditya Dani,, Piyush Nimbalkar, Shardul Mangade, Abhijit Jagdale, Ashish Bhumkar, Priti Biyani, Narita Pandhe, Rushikesh Bhome, Ranjitkumar Burgute, Amey Gaidhani, Swapnil Patil, Pavan Ghuge, Sagar Gavitt, Saurabh Wankhade, Austin Ramos	
	Masters Students – Venkata Ravi Chandra Bandlamudi, Swapnil Pimple, Fiona Jane Britto, Omkar Gawde, Haixin Liu, Xingyu Jin, Jiaan Dai, Jiaqi Zuo, Jiongtao Ye, Sai Kiriti Badam, Xuren Zhou, Arvind Sai Krishnan, Vilas Bhat, Shaobo Guan, Sahar Eslami	
	Ph.D. Students – Timothy Kim, Sanjith Athlur, Kapil Agrawal, Soroush Ghodrati, Suhas Jayaram Subramanya	
SKILLS	Programming Languages - Proficient in C, Python, JavaScript and C++with significant experience in PHP, Java and Shell Scripting.	
	Systems - Proficient in Linux kernel / userspace development, file system architecture and in web application development.	